

COSC 3380-12178 - Team 8: Library Database

Project Info

Github Repo link: <https://github.com/MehrabAzan/Library-Database>

Contributors

Contributor	Main Contributions
Name: Chukwu, David Username: khynoo	● Front-end ○ Pages ■ Initial report ■ Account info
Name: Devkumar Patel Username: devpatel26	● Front-end ● Worked on changing UI for whole front end website(all the pages in the website ● ○ Navbar ○ Notification Icon ■ ○ Sidebar ○ Footer ■ Social Media ○ Pages ■ Account info page ■ Policies page ■ Library Hours Page ■ About us page ■ Policies page ■ Notification page ■ About page ■ Hours page ■ Policies page ○ ● Back-end ○ Website deployment (Frontend with Azure Static App and backend on Render) ○ Database deployment ○ Redesigned Database schema to make triggers work ○ Queries ■ Notificatons (Patron and Staff) ○ Triggers ■ Trigger - 1

	■ Trigger - 2 ■ Non Semantic Triggers - 4 (To support notification bar)
Name: Lin, Evan Username: evan154733	● Front-end ○ on-site notifications ○ Initial navbar ○ Registration functionality ○ Pages ■ Loan ■ Report ■ Fines ● Back-end ○ Reports and their queries ■ Popularity Report ■ User Information ■ Overdue Report ■ Fine Summary Report ○ Queries ■ Registration ■ Log-in ■ Hold (post, patron) ■ Hold (post/get, staff, general) ■ Check-out ■ Loan (get) ○ Authentication ○ Initial SQL testing data file ○ Admin user management
Name: Mehrab Azan Username: MehrabAzan	● Base app & repo ● Front-end ○ Pages ■ Login ■ Forgot Password ■ Account ■ Fines ■ Loans ■ Manage items ● Backend ○ Base setup ○ Queries ■ item modification/deletion ■ Fines (get) ■ Loans (get) ■ User (get) ■ Search query builder ■ Item search query ○ Initial schema ○ Log-in ○ Authentication

	○ Routing ○ Connecting frontend to backend
Name: Morin, Rainer Username: RhythmNotebook	● Front-end ○ Pages ■ Home page ■ Item entry (insertion) ■ Search ■ Registration ■ Staff Registration ○ Components ■ Buttons ■ Dropdown ■ Fine container ■ Input (some pages) ■ Item container (standard/hold/loan) ■ Main page item container ■ Time formatting ○ UI work ■ Major work ● Front page (item containers and sections) ● All account pages (account info, settings, loan/holds, fines, activity) (full page redesign) ● All item management pages (manage items & entry pages) (full page redesign, merging. scrolling.) ● User management (scrolling, redesign) ■ Minor work ● Navbar adjustments (color, bugs, links, display) ● staff pages for loans/holds/fines pages (scrollability, more accessible text colors, search bar changes) ● Reports (more even display) ● Staff signup (field adjustments) ● Back-end ○ Initial schema ○ Queries ■ Item insertion ■ Dropdown item requests ■ User role updating (patron/student/faculty) ■ Main page item queries for each category

Site Info

Host: Azure

Database: Render

Website Link: <https://yellow-desert-027865e1e.7.azurestaticapps.net/>

Due to our setup, the site may take a minute to load up after it has been idle for a while.

Login Information

- Admin
 - Email: admin@library.com
 - Password: password
- Staff
 - Email: staff@library.com
 - Password: password
- Faculty
 - Email: michael@example.com
 - Password: password
- Student
 - Email: emma@example.com
 - Password: password
- Patron
 - Email: john@example.com
 - Password: password

Data Types

Data	Who can add	Who can modify	Who can delete
Books	Staff, Admins	Staff, Admins	Staff, Admins
Audiovisual Media	Staff, Admins	Staff, Admins	Staff, Admins
Equipment	Staff, Admins	Staff, Admins	Staff, Admins
Periodicals	Staff, Admins	Staff, Admins	Staff, Admins
Holds	Patrons, students, faculty, staff, admins	Staff, admins	Patrons, students, faculty, staff, admins
Loans	Patrons, students, faculty, staff, admins	Staff, admins	Patrons, students, faculty, staff, admins
Fines	Staff, admins	Patrons, students, faculty, staff, admins	N/A
Patron Accounts	Anyone	Patrons*, students*, faculty*, staff, admins (*most account info)	Staff, admins

Staff account	Admins	Staff, admins	Admins
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User Roles

Role	Explanation	Abilities
Patron	Standard account for anyone who signs up as a library user	Can search for, hold, and loan items (books, audiovisual media, periodicals, equipment) and pay fines. Has due dates of 2 weeks and fines of 0.25/day.
Student	Role for students as verified and elevated by staff or admins.	Has extended due dates (3 weeks) and lower fines (0.20/day) compared to above.
Faculty	Role for faculty as verified and elevated by staff or admins.	Has extended due dates (weeks) and lower fines (0.10/day) compared to above.
Staff	General role for library staff.	Can add, modify, and delete library items (books, periodicals, audiovisual media, equipment). Can waive and alter patron fines. Can manage holds, loans, and checking out items.
Admin	Staff role with elevated privileges for more management-oriented needs.	Has staff privileges. Can view and adjust various reports. Can sign up staff or admin accounts. Can manage user accounts and their information.

Semantic Constraints

Trigger	Purpose	Code
Notify patron when hold is ready	Automatically sends a notification and logs the event when a hold status changes to "Ready for Pickup" (code 2).	<pre> CREATE TRIGGER notify_hold_ready AFTER UPDATE ON holds FOR EACH ROW BEGIN IF OLD.hold_status_code != 2 AND NEW.hold_status_code = 2 THEN INSERT INTO notifications (patron_id, notification_type, message, link_data) </pre>

		<pre> VALUES (NEW.patron_id, 'HOLD READY', CONCAT('Good News! Your Book is ready for pickup!'), JSON_OBJECT('hold_id', NEW.hold_id, 'item_id', NEW.item_id)); -- Log the activity INSERT INTO activity_logs (actor_type, actor_id, action_type, hold_id, description, created_at) VALUES ('staff', IFNULL(@current_staff_id, 0), 'HOLD_READY_NOTIFICATION', NEW.hold_id, CONCAT('Notification sent to patron ', NEW.patron_id), NOW()); END IF; END\$\$ </pre>
Charge overdue fine on return	Calculates and applies a fine to a patron's account if an item is returned after the due date, based on their specific patron role rate.	<pre> CREATE TRIGGER notify_fine_created_enhanced AFTER INSERT ON fines FOR EACH ROW BEGIN DECLARE days_late INT; DECLARE daily_fine_amount FLOAT; DECLARE notification_msg VARCHAR(500); DECLARE notification_type VARCHAR(50); -- Check if this is an overdue fine (has loan_id) IF NEW.loan_id IS NOT NULL THEN -- Get loan details for calculation SELECT DATEDIFF(l.return_date, l.loan_due_date), pr.fine INTO days_late, daily_fine_amount FROM loans l JOIN patron_roles pr ON pr.patron_role_code = l.patron_role_code WHERE l.loan_id = NEW.loan_id LIMIT 1; -- Create detailed overdue notification SET notification_type = 'OVERDUE FINE'; SET notification_msg = CONCAT('Late Return Fee: You were charged \$', ROUND(daily_fine_amount, 2), ' for today. Total Amount due: \$', ROUND(NEW.fine_amount, 2)); ELSE -- Regular fine (not overdue-related) SET notification_type = 'FINE CREATED'; END IF; END; </pre>

		<pre> SET notification_msg = CONCAT('A fine of \$', ROUND(NEW.fine_amount, 2), ' has been applied to your account.'); END IF; -- Create notification INSERT INTO notifications (patron_id, notification_type, message, link_data) VALUES (NEW.patron_id, notification_type, notification_msg, JSON_OBJECT('fine_id', NEW.fine_id, 'amount', NEW.fine_amount, 'loan_id', NEW.loan_id)); -- Log the activity INSERT INTO activity_logs (actor_type, actor_id, action_type, fine_id, target_patron_id, description, created_at) VALUES ('system', 0, 'FINE_NOTIFICATION', NEW.fine_id, NEW.patron_id, notification_msg, NOW()); END\$\$ </pre>
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Queries and Reports

Report	Purpose	Displays	User Adjustments
Popularity	View most popular items in the library, including different categories. (etc....).	Overall: - Total loan count - Unique items borrowed - Most popular category - Most popular genre - Top borrowed item (and its loans) - Average loans per item Detail: - Loan count - Loan history	- Start/end date - Category - Genre - Text in attribute - Sort by attribute - Sort direction (ascending/descending)
User Overview	Look up a patron by ID and view their holds, loans, and fines. (etc....).	Overall: - Total patrons - Total loans history - Current loans - Active holds - Outstanding balance	- Start/end date - Activity - Search by name or id - Sorting by loans/holds/fines/

		(total) • Patrons with balance • Patrons with activity • Most active patron Detail: • total/current/returned loans • total/active holds • Fine records • Outstanding/paid balances • Last loan/hold/fine/activity dates	payments/activity date/name/id • Sort direction (ascending/descending)
Operation	View all overdue loans, the patron responsible, overdue days, and current fine totals. (etc....)	Overall: • Total overdue loans • Total overdue patrons • Average days overdue • Maximum days overdue • Estimated outstanding fines (aggregate) • Most overdue item Detail: • Item name/id and patron name/id • Borrow/due date • Days overdue • Daily fine for user • Estimated fine • Return/mark as lost actions	• Start/end date • Item category • Minimum days overdue • Search by patron name/id, item name/id, item category, or creator • Sort by days overdue, due date, borrow date, daily fine, patron name/id, item title/id, category, creator, or loan id • Sort direction (ascending / descending)
Fine Summary	Report of total fines, paid amounts/remaining balances. (etc....).	Overall: • Total fine records • Total fine amount • Total paid amount • Total remaining amount • Average fine • Average days overdue • Overdue (and aggregate overdue fines) • Returned but unpaid (and aggregate fines in this category) • Paid (and aggregate paid fines) • Waived (and aggregate waived fines)	• Start/end date • Status (overdue/returned but unpaid/paid/waived) • Minimum days overdue • Search by name/id/book name/creator/status/fine id • Sort by remaining amount/total fine/paid amount/days

			overdue/patron name/book name/due date/status/fine id • Sort direction (ascending / descending)
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